Case Studies in Operations Management

Mathematical Model

Business and Logistics Plan for "Fresh Veggie" (FV)

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Date:
April 9, 2026

1 Model Assumptions

2 Sets

- DC: set of candidate distribution centers.
- DS: set of candidate dark stores.
- $MDCP$: set of candidate parking locations for mobile distribution centers.
- $F = DC \cup DS \cup MDCP$: set of all candidate facilities in the network.
- P: set of ports:
$$P = \{\text{San Antonio}, \text{Valparaíso}\}$$
- A: set of airports:
$$A = \{\text{Arturo Merino Benítez International Airport}\}$$
- $E = P \cup A$: set of international entry points to Chile.
- WM: set of wholesale markets:
$$WM = \{\text{La Vega Central}, \text{Lo Valledor}\}$$
- S^{imp}: set of imported supplier countries:
$$S^{imp} = \{\text{US}, \text{Mexico}, \text{Peru}, \text{Ecuador}, \text{Bolivia}, \text{Argentina}, \text{Spain}\}$$
- S^{cl}: set of domestic supplier regions in Chile:
$$S^{cl} = \{\text{Chile-Scl}, \text{Chile-Vap}, \text{Chile-Maul}, \text{Chile-Coq}\}$$
- $S = S^{imp} \cup S^{cl}$: set of all supplier origins, where
$$S^{imp} \cap S^{cl} = \varnothing$$
- K: set of products:
$$K = \{\text{Roma Tomatoes}, \text{Cauliflower}, \text{Broccoli}, \text{Asparagus}, \text{Green Bell Pepper}\}$$
- T: set of weeks in the planning horizon:
$$T = \{1, 2, \ldots, 52\}$$
- R: set of transportation modes:
$$R = \{\text{sea}, \text{air}, \text{truck}\}$$
- $\mathcal{Z}$: set of all feasible facility configurations, where each $z \in \mathcal{Z}$ is a binary vector in $\{0, 1\}^{|F|}$, excluding the null configuration.

3 Parameters

3.1 Demand and Market Parameters

3.2 Revenue Parameters

3.3 Sourcing Parameters

3.4 Distance Parameters

- $D_{i,j}^{DC,DS}$: distance between distribution center $i \in DC$ and dark store $j \in DS$, expressed in km.
- $D_{i,m}^{DC,MDCP}$: distance between distribution center $i \in DC$ and mobile distribution center parking location $m \in MDCP$, expressed in km.
- $D_{e,i}^{E,DC}$: distance between international entry point $e \in E$ and distribution center $i \in DC$, expressed in km.
- $D_{s,i}^{S^{cl},DC}$: distance between domestic supplier $s \in S^{cl}$ and distribution center $i \in DC$, expressed in km.
- $D_{m,n}^{MDCP,MDCP}$: distance between mobile distribution center parking locations $m \in MDCP$ and $n \in MDCP$, expressed in km.

3.5 Time Parameters

- $T_{i,j}^{DC,DS}$: travel time between distribution center $i \in DC$ and dark store $j \in DS$, expressed in hours.
- $T_{i,m}^{DC,MDCP}$: travel time between distribution center $i \in DC$ and mobile distribution center parking location $m \in MDCP$, expressed in hours.
- $T_{p,i}^{P,DC}$: travel time between port $p \in P$ and distribution center $i \in DC$, expressed in hours.
- $T_{a,i}^{A,DC}$: travel time between airport $a \in A$ and distribution center $i \in DC$, expressed in hours.
- $T_{e,i}^{E,DC}$: travel time between international entry point $e \in E$ and distribution center $i \in DC$, expressed in hours.
- $T_{s,i}^{S^{cl},DC}$: travel time between domestic supplier $s \in S^{cl}$ and distribution center $i \in DC$, expressed in hours.
- $T_{m,n}^{MDCP,MDCP}$: travel time between mobile distribution center parking locations $m \in MDCP$ and $n \in MDCP$, expressed in hours.
- $T_{s,e}^{S^{imp},E}$: travel time between imported supplier $s \in S^{imp}$ and international entry point $e \in E$, expressed in hours.

3.6 Transportation Parameters

3.7 Network Design Parameters

3.8 Inventory Parameters

3.9 Perishability Parameters

3.10 Service Level Parameters

3.11 Cost Parameters

- $C^{pur,imp}_{k,s,e}$: purchasing cost per kilogram of product $k \in K$ acquired from imported supplier $s \in S^{imp}$ through international entry point $e \in E$, expressed in \$/kg.
- $C^{pur,cl}_{k,s}$: purchasing cost per kilogram of product $k \in K$ acquired from domestic supplier $s \in S^{cl}$, expressed in \$/kg.

3.12 Stochastic Parameters

4 Decision Variables

5 Objective Function

6 Constraints

7 Appendices